



# Analysis of the influence of cathode recirculation strategy on proton exchange membrane fuel cell from internal polarization and external characteristics perspectives

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## HIGHLIGHTS

- A systematic study of cathode recirculation on PEMFC under low-load conditions.
- The polarization process at various recirculation strategies are identified by DRT.
- Cathode circulation has positive prospects in vehicle self-humidifying PEMFC system.

## ARTICLE INFO

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## ABSTRACT

Cathode recirculation provides a new solution to the accelerated degradation of proton exchange membrane fuel cells (PEMFCs) due to low humidity and high potential at low load conditions. Analyzing the internal impact mechanisms is basis for a more appropriate use of this technology. This paper presents a systematic investigation of the mechanisms by which key operating parameters of the cathode recirculation, including current density, cathode recirculation ratio, and stoichiometry of the fresh air path, affect the internal and external properties of PEMFCs. The voltage suppression effect and humidification effect under different cathode recirculation strategies are first analyzed. Then, as a novelty, the distribution of relaxation times (DRT) method is used to deconvolve electrochemical impedance spectra and qualitatively resolve the polarization impedance under different strategies. DRT is enabled to separate up to five polarization processes with different time constants and assign them to oxygen diffusion, oxygen reduction reaction and proton transport processes affected by the cathode recirculation. Finally, DRT quantified the change in ohmic impedance due to cathode recirculation and confirmed the effect of external humidity changes. This work provides a more in-depth insight into internal mechanisms of cathode recirculation in PEMFCs, and especially provides a comprehensive theoretical basis for the development of idle controllers for automotive PEMFCs in terms of improved durability.

## 1. Introduction

Proton exchange membrane fuel cells (PEMFCs) in recent years have received considerable attention in recent years due to its excellent performance such as high efficiency, high energy density, low noise and no pollution, and is considered as one of the most promising solutions for on-board power systems [1–3]. However, variable and complex operating conditions of automobiles lead to key materials of PEMFC (e.g. catalysts and carbon supports) rapid and irreversible degradation, which consequently makes the durability of PEMFC a major challenge for the

widespread commercialization fuel cell vehicles (FCV) today [4–6].

Low relative humidity and high potential are two main problems in PEMFC system under low load conditions, especially in the state-of-the-art self-humidifying system, where humidifying absolutely relies on the self-produced water and hydrogen recirculation mechanisms, and where the amount of water produced is usually insufficient [7]. Meanwhile, high potential under low load conditions causes key materials rapid decay [8]. All of these could irreversibly and rapidly reduce the durability of the PEMFC. Therefore, it is essential to research and develop optimization strategies to improve the service life of PEMFC as long as possible. Equipping PEMFC with cathode recirculation draws on

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| Abbreviations |                                        | Symbol       |                                           |
|---------------|----------------------------------------|--------------|-------------------------------------------|
| PEMFC         | Proton exchange membrane fuel cell     | $i$          | Current density                           |
| PEM           | Proton exchange membrane               | $\varphi$    | Cathode recirculation ratio               |
| EGR           | Exhaust gas recirculation              | $\lambda_0$  | Fresh air path stoichiometry              |
| EIS           | Electrochemical impedance spectroscopy | $R_0$        | Ohmic resistance                          |
| AC            | Alternating current                    | $R_{pol}$    | Polarization impedanc                     |
| ECM           | Equivalent circuit model               | $\lambda$    | Regularization coefficient                |
| RL            | Resistor-inductor                      | $W$          | Mass flow                                 |
| DRT           | Distribution of relaxation times       | $\tau$       | Time constant                             |
| MEA           | Membrane electrode assembly            | $\omega$     | Angle frequency                           |
| Pt            | Platinum                               | $g(\tau)$    | Distribution function of relaxation times |
| PTFE          | Polytetrafluoroethylene                | $Z_{DRT}(f)$ | DRT fitted impedance                      |
| DC            | Direct current                         | $Z_{Re}$     | Real part of the impedance                |
| ORR           | Oxygen reduction reaction              | $Z_{Im}$     | Imaginary part of impedance               |
| KK            | Kramers-Kronig                         | $\Delta Re$  | Residual of the real part                 |
| RH            | Relative humidity                      | $\Delta Im$  | Residual of the imaginary part            |
| T             | Temperature                            |              |                                           |

the anode recirculation principle of the PEMFC and the exhaust gas recirculation (EGR) principle of the internal combustion engine, which could effectively address the above-discussed two problems by reasonably designing the cathode recirculation strategy when PEMFCs suffer from low variable loads conditions. Cathode recirculation involves reintroducing the exhaust gases from the cathode outlet into the cathode inlet, where they mix with fresh air. The optimization effects of this process can be explained from two perspectives. Firstly, a significant amount of water vapor in the exhaust gases humidifies the dry fresh air, thereby increasing the intake humidity under low-load conditions. Secondly, the concentration of oxygen in the mixed intake air decreases as a result of mixing the exhaust gases with fresh air. Since the level of oxygen concentration directly determines the magnitude of the PEMFC voltage, high voltage under low-load conditions is suppressed. Yang et al. [9] in their study firstly report a cascaded cell group architecture that allows cathode reaction gas of the previous cell to humidify the next one. Kim [10] in their study report the effect of humidification by cathode exhaust recirculation and results show that cathode recirculation effectively improve the ability of self-humidifying, although slightly reduce performance in normal conditions compare with traditional external humidifier. Rodosik et al. [11] investigate the impact of humidification by cathode exhaust gases recirculation on a 5 kW PEMFC system for automotive applications, and results show that medium and high recirculation ratio is more suitable for long term performance stability. Zhang et al. [12] analyze the effect of operation parameters on water transport of PEMFC through building a cathode recirculation model. They declare that the maximum relative humidity reach to 60% at medium current density. Liu et al. [13] study a dual-ejector recirculation PEMFC stack performance, and results show that dual-cycle mode alleviate the gas shortage problem during dynamic loading compared to dead anode and cathode mode. Zhao et al. [14] investigate dynamic characteristics of dual-cycle PEMFC system, they find cathode recirculation achieve more even distribution of water in membrane, and oxygen concentration difference between inlet and outlet is smaller. Liu et al. [15] studies the dynamic performance of cathode recirculation at high current density conditions, and find that the liquid water can be removed promptly, thereby reducing the risk of flooding and improve the consistency of the single-cells. Liu et al. [16] further explore the control issue of cathode recirculation on high power on-board PEMFC system, they design a high effective adaptive controller achieve accuracy voltage adjustment through control pump speed. Indeed, although numerous studies recently explore cathode recirculation's performances in PEMFC. However, it still lack effective and deep experimental verification. There is still a lack of a valid explanation of the internal

dynamics of PEMFC equipped with cathode recirculation, especially from the combination perspective of internal polarization and external characteristics.

Electrochemical impedance spectroscopy (EIS) is a suitable in-situ technique for recording and evaluating polarization processes with individual time constants [17]. EIS have so far been widely applied to identify different loss process in PEMFC due to its ease of offline and online application attributes. It is convenient to characterize internal loss changes in PEMFC caused by different operating parameters and components such as humidity, current density, stoichiometry, and membrane and electrolyte resistance and so on [18]. Yuan et al. [19] detailed review the application of AC impedance technique on PEM fuel cell, especially in the in situ impedance measurements. Tang et al. [20] further review recent works in the use of EIS for the measurement, monitoring, diagnosis and optimization of PEMFC performance. Hong et al. [21] propose a novel approach to online AC impedance measurement at high frequency of PEM fuel cell stack for the accurate water content estimation. Zhang et al. [22] establishes a framework for online electrochemical diagnosis of on-board PEMFC, and results show that optimized electrochemical impedance spectroscopy based on dynamic multi-frequency analysis is the best diagnostic method for online application.

The main method nowadays for analyzing the impedance spectrum information of PEMFC is the equivalent circuit model (ECM) using complex nonlinear least square fit approximation [23]. This model typically consist of a series of electronic components such as resistances, capacitors, and inductors. Through assigning appropriate values to the electrical components in order to fit experiment data and quantitatively analyze impedance more accurately [24]. Niya et al. [25] review the applications of EIS in PEM fuel cell modeling and diagnosis, and discuss the critical points that can be misinterpreted in the ECM method in the field of PEMFC. Pivac et al. [26] discuss the inductive phenomena at low frequencies of PEMFC using equivalent circuit models. Yuan et al. [27] investigate the effect of various operating conditions on impedance spectra of low temperature fuel cell, and quantitatively analyze polarization process using ECM method. Xiao et al. [28] explore the impact of drying faults on PEMFC under different operating condition, and the EIS results are identified parametrically using the RL equivalent circuit model to obtain ohmic, charge transfer and mass transfer impedance values.

However, ECM analysis method requires a priori knowledge of the physical electrochemical processes of the object [26]. There is an inevitable challenge: the EIS model of the object is not always accurately known, and the model itself may change in response to internal or

external factors. Moreover, ECM struggles with evaluating polarization processes with close time constants (processes overlapping in the frequency domain), which easily leads to incorrect model assumptions and fitting results [29,30]. In this light, a model-free, without prior knowledge method, distribution of relaxation times (DRT), is proposed and is widely used for analyzing EIS data. DRT method have obtain successful experiences in field of lithium-ion batteries, super capacitors and solid oxide fuel cells [31–33]. Recently, DRT method also realizes successful transfer application on polymer electrolyte membrane fuel cells. Weiß et al. [34] analyze the impedance spectrum of a HT-PEMFC, and allocate the DRT peaks to loss mechanisms inside the fuel cells. Bevilacqua et al. [35], explore the impact of anode operating conditions on polarization loss of a HT-PEMFC using DRT method. Heinzmann et al. [36] reveal anode contributions that usually are hidden by the dominating cathode contributions in loss process through DRT deconvolution. Kwon et al. [37] used RT-based impedance analysis method to determine the degradation rate of carbon corrosion reaction of PEMFC. Zhao et al. [38] measures the impedance of the PEMFC stack using multi-channel impedance test system, and use the DRT to guide the establishment of the ECM and analyze its local sensitivities to working conditions.

For PEMFC, the internal polarization process is generally dominated by the cathode side, and although the anode also contributes to the extent, it is usually neglected. The focus of this study is on the effect of the cathode recirculation strategy on the PEMFC, and therefore it is extremely essential to analyze the changes in internal dynamics. The main contents and contributions of this study are as follows:

- 1) Taking the rule that the total inlet mass flow rate at the cathode side is constant, the cathode recirculation ratio is changed to achieve the effects of humidification and voltage suppression under low load conditions.

- 2) A  $5 \times 5$  single cell is widely tested on a fuel cell test bench to investigate the effect of cathode recirculation strategy on PEMFC's performance, including air stoichiometry, cathode recirculation ratio, and current density.
- 3) Combined with external characterization, the DRT method is used to quantify the internal polarization process of PEMFC under different cathode strategies to provide more comprehensive parsing and insights.

## 2. Experiment setup and procedure

### 2.1. Tested MEA and experiment bench

A  $25 \text{ cm}^2$  commercial membrane assembly (MEA) is used in this study, the Pt loading of the anode is  $0.1 \text{ mg/cm}^2$  and the Pt loading of the cathode is  $0.3 \text{ mg/cm}^2$ . The flow channels at both sides are serpentine channels in parallel wherein the PTFE gaskets (thickness:  $100 \mu\text{m}$ ) are used for sealing. All experiments are conducted on the Greenlight G20 test station, its schematic diagram and actual test bench are shown in Fig. 1. The pure hydrogen is supplied through hydrogen storage tanks, and the compressed air is provided by a compressor. The mass flow rate and inlet gas pressure for the anode and cathode were determined by mass flow controllers and backpressure valves, respectively. The inlet gas can directly be supplied to the single cell or be humidified by the bubble humidifier. The temperature of cell is adjusted by the water bath, and a thermocouple placed at the cathode provides the real-time temperature measurement, which could be regarded as whole-cell temperature during the operation. Additionally, the electrochemical workstation, Gamry Interface 5000E, is used to measure EIS. The frequency range in this study is from  $10 \text{ kHz}$  to  $0.1 \text{ kHz}$ , and the amplitude is picked to be 8% of DC load to achieve a linear measure.

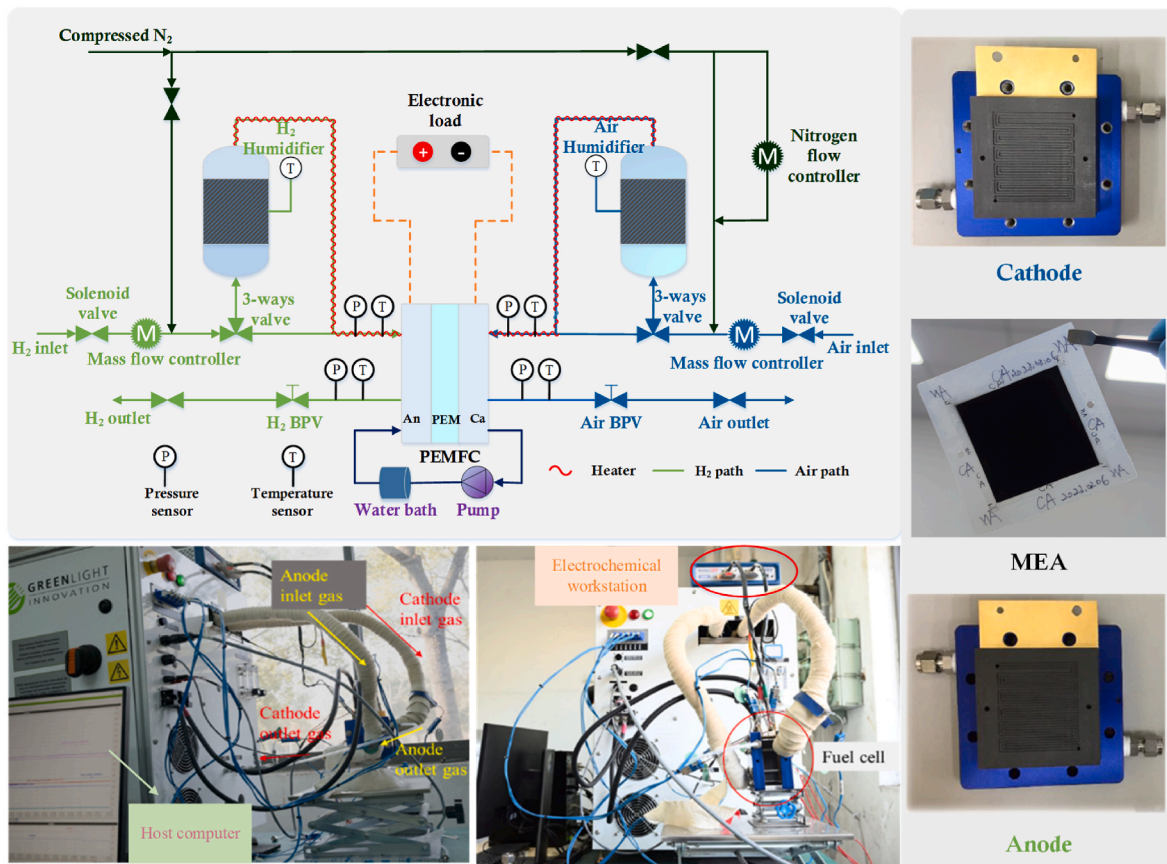


Fig. 1. Schematic, physical and MEA pictures of the experimental single-cell test station.

## 2.2. Experiment procedures and activation

In this study, the total intake mass flow in cathode side is kept constant. This means that a fresh air path stoichiometry ( $\lambda_0$ ) is determined when recirculation is not applied, and its corresponding air mass flow is the baseline. Varying inlet air humidity and oxygen mass flow by adjusting the cathode recirculation ratio ( $\varphi$ ). The cathode recirculation ratio is defined as shown in Equation (1). Since the air for the test bench is supplied in a straight line way and the exhaust gases are discharged directly to the atmosphere. Therefore, this work adopt a gas supply way that simulates the inlet gas conditions of cathode recirculation by strict theoretical calculations. Three main assumptions are made: 1) the vapor in the outlet gas is saturated, 2) only oxygen, nitrogen and vapor are recirculated, and 3) all gases are treated as ideal gases and follow the ideal gas law. Specifically, the ratio of oxygen, nitrogen, and water vapor at the inlet under different  $\varphi$  is recalculated according to the law of conservation. There are two gas supply line: one feeds the dry air mass flow corresponding to the total oxygen mass flow, the other feeds additional nitrogen mass flow, and the humidity (water vapor mass flow) is regulated by a humidifier under corresponding dew point temperature. The dew point temperature of humidifier are calculated as Equations (2) and (3). Cathode relative humidity is calculated as Equations (2)–(4), the amount of dry air and additional nitrogen is calculated by equations (5) and (6). More detailed derivation of Eq. (1) to Eq. (6) can be accessed in the Supplementary Material.

increase, this is because the accumulation of liquid water in the channel at high current densities challenges the transport of oxygen, thus increasing the mass transfer resistance. Moreover, the radius of the impedance arc in the medium-frequency region first decreases and then increases. One possible reason is that the low catalyst activity and low oxygen concentration at small current density leads to high charge transfer resistance, which improves with increasing current density. However, as the current density continue to increase, the reduction of active site caused by flooding and the rapid consumption of oxygen increase the charge transfer resistance again.

$$\varphi = \frac{W_{\text{cycle}}}{W_{\text{fresh}}} \quad (1)$$

$$p_{\text{sat}}(T_{\text{dew}}) = RH_{\text{inlet}} p_{\text{sat}}(T_{\text{st}}) \quad (2)$$

$$\log_{10} p_{\text{sat}}(T_{\text{st}}) = -2.1794 + 2.95 \times 10^{-2}(T_{\text{st}} - 273.15) - 9.1837 \times 10^{-5}(T_{\text{st}} - 273.15)^2 + 1.4454 \times 10^{-7}(T_{\text{st}} - 273.15)^3 \quad (3)$$

$$RH_{\text{inlet}} = \begin{cases} 0 & \text{no recirculation} \\ \frac{\varphi}{1 + \varphi} & \text{recirculation} \end{cases} \quad (4)$$

$$W_{\text{normal,air}} = \frac{nI \cdot \lambda_0 \cdot (0.21M_{O_2} + 0.79M_{N_2})}{4F \cdot 0.21 \cdot (1 + \varphi)} \cdot \left[ 1 + \frac{\lambda_0 - 1 - \varphi}{\left( \frac{0.79M_{N_2}}{0.21M_{O_2}} + 1 \right) \cdot \lambda_0 - 1 - \varphi} \cdot \varphi \cdot \frac{\frac{0.79M_{N_2}}{0.21M_{O_2}} + 1}{1 + \varphi} \right] \quad (5)$$

Since this paper is concerned with the effect of cathode recirculation on PEMFC under low load conditions, the current density is set in the region of low current density. The detailed experiment conditions in this work is shown in Table 1. Before conducting experiment, the fuel cell is activated for 4 h, with the current density set to 1.8 A/cm<sup>2</sup> and 100% humidification to ensure stable performance output during the experiment. The initial electrochemical performance after activation is shown in Fig. 2. In Fig. 2(a), we measure the I-V and power curves of the fuel cell in loaded and reduced load modes, respectively. The maximum voltage error is within 5% in both modes and the maximum power density over 1W/cm<sup>2</sup>, which indicates the stability of the fuel cell in current situation. Fig. 2(b) shows the impedance information, we measure the EIS when the current density is 0.3A/cm<sup>2</sup>, 0.9A/cm<sup>2</sup>, 1.3A/cm<sup>2</sup>, 1.6A/cm<sup>2</sup> and 2.0 A/cm<sup>2</sup>. The entire electrochemical impedance spectrum consists of ohmic impedance ( $R_0$ ) and polarization impedance ( $R_{\text{pol}}$ ). Inductive phenomena (the part below zero on the y-axis in Fig. 2 (b)) is observed in the high frequency band and the low frequency band. The inductive features at high frequencies do not represent the physical and chemistry of the PEMFC, and its occurrence can be explained by the cables and wires of the test system [39]. The inductive features at low frequencies occur for a variety of reasons. The most widespread explanations are that it is related to the side reactions with intermediate species, the water transport properties and carbon monoxide poisoning, but so far these explanations seem to remain ambiguous [26]. In the subsequent discussion of DRT in this study, we ignore the inductive information irrelevant to capacitive dependent dynamics.

$R_0$  remain essentially constant at different current density. Generally, ohmic loss is principally determined by the ionic resistance, 100% humidification in test process leads to membrane is fully wet so that the proton transfer resistances is almost unchanged. As the current density increases, the radius of the impedance arc in the low-frequency region

$$W_{\text{add,N}_2} = \frac{0.79M_{N_2}}{0.21M_{O_2}} \cdot \frac{nI \cdot \lambda_0 \cdot (0.21M_{O_2} + 0.79M_{N_2}) \cdot \varphi}{4F \cdot 0.21 \cdot (1 + \varphi) \cdot \left[ \left( \frac{0.79M_{N_2}}{0.21M_{O_2}} + 1 \right) \cdot \lambda_0 - 1 - \varphi \right]} \quad (6)$$

Where,  $W_{\text{cycle}}$  is the circulated gas mass flow,  $W_{\text{fresh}}$  is the fresh gas mass flow,  $p_{\text{sat}}(T)$  is the saturated vapor pressure under temperature  $T$ ,  $RH_{\text{inlet}}$  is the relative humidity at cathode inlet,  $W_{\text{normal,air}}$  the dry air mass flow corresponding to the total oxygen mass flow,  $W_{\text{add,N}_2}$  additional nitrogen mass flow,  $\lambda_0$  is the excess oxygen ratio of the fresh air path.

## 3. Analysis method

### 3.1. Kramers-Kronig valid

In order to correct physical quantities from impedance spectra, the

**Table 1**

Operating conditions in this study.

| Parameters                                  | Values                   | Unit              |
|---------------------------------------------|--------------------------|-------------------|
| Current density                             | 0, 0.05, 0.1, 0.15, 0.2  | A/cm <sup>2</sup> |
| Cathode stoichiometry without recirculation | 2, 3, 4, 5               |                   |
| Cathode recirculation ratio                 | 0, 0.2, 0.4, 0.6, 0.8, 1 | \                 |
| Cell temperature                            | 60                       | °C                |
| Cathode/anode back pressure                 | 0.2                      | Mpa               |
| Hydrogen mass flow                          | 0.07                     | SLPM              |
| Cathode/anode inlet temperature             | 60                       | °C                |
| Relative humidity at anode                  | 100                      | %                 |
| Flow mode of Cathode and anode              | Counter-flow             | \                 |

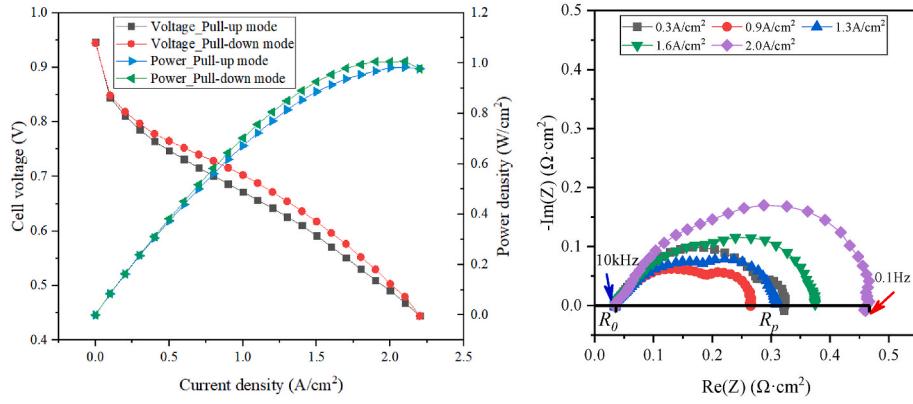


Fig. 2. The initial electrochemical performance after activation. (a) polarization curve, (b) EIS.

spectra must exhibit very high quality and must not contain errors. An effective way to ensure the validity of EIS data is to perform a Kramers-Kronig transformation [40]. The KK test relate the real part of the impedance spectrum of a linear, time-invariant and causal system to its imaginary part and vice versa [41]. The relevant equations are shown in Equations (7) and (8).

$$Z_{Re}(\omega) = \frac{2}{\pi} \int_0^{\infty} \frac{\omega' \cdot Z_{Im}(\omega')}{\omega^2 - \omega'^2} d\omega' \quad (7)$$

$$Z_{Im}(\omega) = -\frac{2}{\pi} \int_0^{\infty} \frac{\omega' \cdot Z_{Re}(\omega')}{\omega^2 - \omega'^2} d\omega' \quad (8)$$

Where,  $Z_{Re}(\omega)$  represents the real part of the impedance,  $Z_{Im}(\omega)$  is the imaginary part,  $\omega$  is the angle frequency. A sufficiently general model is used fitting impedance spectra where contains a series connection of an ohmic resistor and RC-elements. In order to judge the reproducibility of a measured impedance spectrum by the fitted equivalent circuit model, the residual  $\Delta$  is calculated:

$$\Delta Re(\omega) = \frac{Z_{Re}(\omega) - \hat{Z}_{Re}(\omega)}{|Z(\omega)|} \quad (9)$$

$$\Delta Im(\omega) = \frac{Z_{Im}(\omega) - \hat{Z}_{Im}(\omega)}{|Z(\omega)|} \quad (10)$$

Where  $|Z(\omega)|$  is the absolute value of the measured impedance and  $\hat{Z}(\omega)$  is fitted impedance. In this work, KK tool [42] is used to prove the effectiveness of EIS data. Generally, if the largest residual is less than 1%, it is considered to pass the KK validation. Fig. 3(a) shows the impedance spectrum of a working condition point in this experiment, and the illustrated frequency range is from 0.1 Hz to 10 kHz. Fig. 3(b) shows the relative residual between Kramers-Kronig model and experiment EIS data. The relative residual in the evaluated frequency range is less than 1%. Based on this criteria, all EIS data in this study satisfy the KK test validity.

### 3.2. DRT theory

The DRT method is more adept at quantitative analysis of polarization processes, and the relaxation time distribution of electrochemical systems is extracted by the inverse convolution technique to achieve the separation and resolution of highly overlapping physicochemical processes in EIS. The relationship between the complex impedance and the relaxation time distribution function is shown in Equation (11) [43]:

$$\begin{cases} Z(\omega) = R_0 + Z_{pol}(\omega) = R_0 + R_{pol} \int_0^{\infty} \frac{g(\tau)}{1 + j\omega\tau} d\tau \\ \text{where, } \int_0^{\infty} g(\tau) d\tau = 1 \end{cases} \quad (11)$$

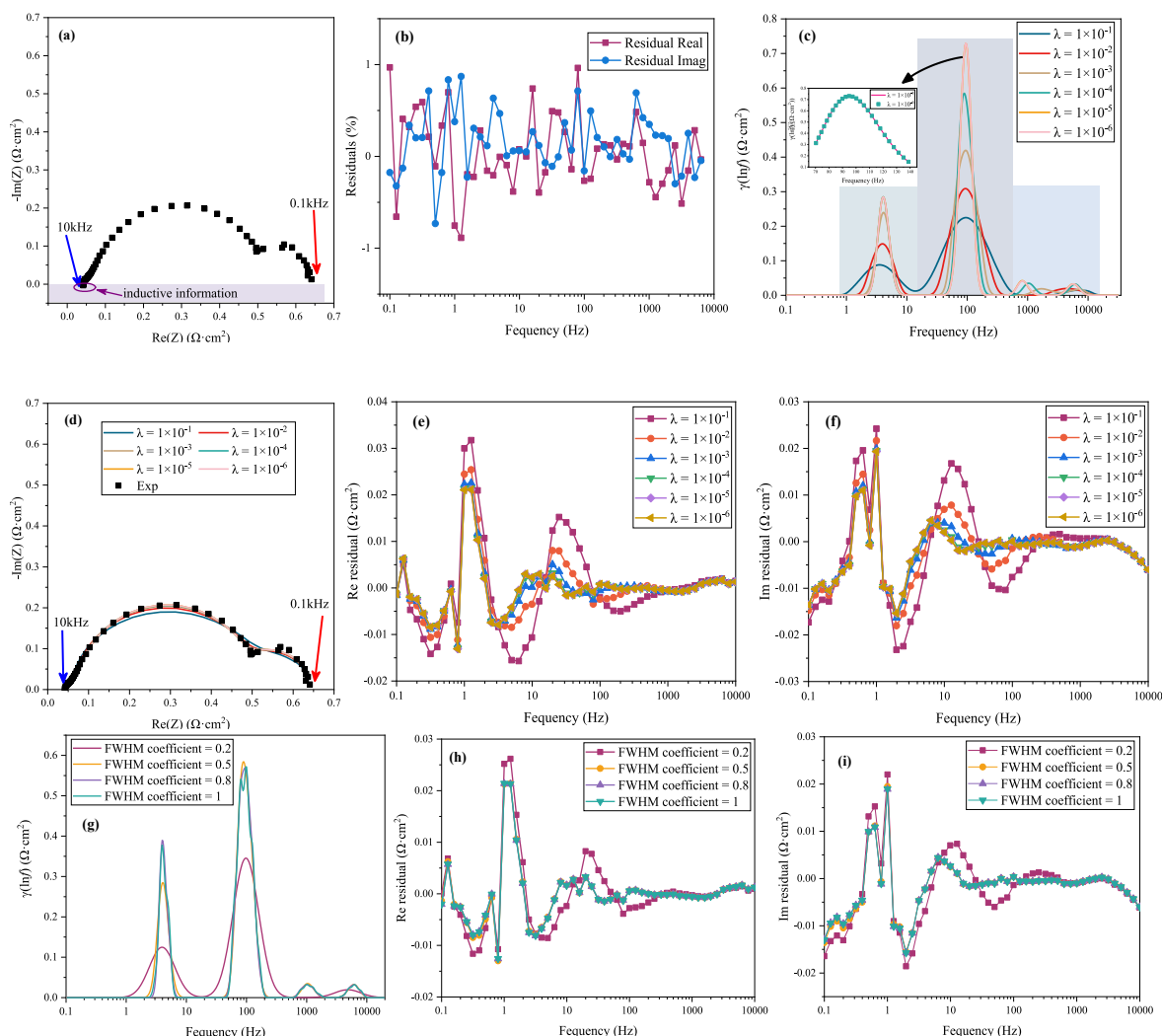
Where,  $R_0$  represents the ohmic resistance and  $R_{pol}$  is the polarization resistance.  $\tau$  is the characteristic time constant of the relaxation,  $g(\tau)$  is the distribution function of relaxation times,  $j$  is a plural unit,  $\omega$  is an angular frequency. Generally, since the frequency data is often collected on the logarithmic scale with a given number frequencies per decade, the complex impedance fitted by the DRT model can be rewritten as follows [42]:

$$Z_{DRT}(f) = R_0 + R_{pol}(f) = R_0 + \int_0^{\infty} \frac{\gamma(\ln \tau)}{1 + i2\pi f\tau} d\ln \tau \quad (12)$$

Where  $\gamma(\ln \tau) = \tau g(\tau)$ , it's a suitable function that describes the time relaxation characteristics of the electrochemical system studied.  $Z_{DRT}(f)$  is the fitted impedance of DRT model and  $f$  is the frequency. In this study, we use DRTtools an open source software developed by the Ciucci's research group [44,45], to perform DRT analysis of the measured impedance data.

### 3.3. DRT solution setup

The DRT method can decompose each contribution to the impedance spectrum and show each process by the peaks in the DRT spectrum. However, overlap may sometimes occur. Commonly, the number and shape of identifiable peaks highly depend on the regularization parameter  $\lambda$ . Therefore, it is essential to assess the proper value of  $\lambda$  to obtain accuracy and meaningful distribution functions. Fig. 3(c) shows the peak distribution of DRT under different  $\lambda$  that changes from  $10^{-1}$  to  $10^{-6}$ . It can be found that three peak are identified when  $\lambda = 10^{-1}$  and  $10^{-2}$ , and with the decrease of  $\lambda$ , the number of peaks increase. It is clear that all polarization processes can be broken down by decreasing the  $\lambda$ . In contrast, higher  $\lambda$  may ignore some process which have close time constant. As the  $\lambda$  continue to decrease to too small value such as  $\lambda = 10^{-5}$  and  $10^{-6}$ , the number of peaks remains constant while the distribution of peaks becomes sharper, which increase the risk of oscillation. The reconstructed Nyquist plots and corresponding residual (calculated by Eqs 10 and 11) are shown in Fig. 3(d-f). It is obvious that the residuals decrease with lower  $\lambda$  but do not become better for  $\lambda$  values below  $10^{-4}$ . Therefore, the optimal  $\lambda$  value is set to  $10^{-4}$  in this study. Moreover, the full width at half maximum (FWHM) coefficient of the Gaussian discretization is another critical parameter. Fig. 3(g) shows the sensitivity of the DRT analysis results to the FWHM coefficient. It can be observed that when the FWHM coefficient is too small (0.2), the DRT results are overly smooth and neglect certain peaks. Conversely, when



**Fig. 3.** Analysis of impedance data. (a) Nyquist plot of impedance data, (b) Kramers-Kronig test residuals, (c) DRT results against various  $\lambda$ , (d) Reconstructed Nyquist plots against various  $\lambda$ , (e) Corresponding residuals of real parts, (f) Corresponding residuals of imaginary parts, (g) DRT results against various FWHM coefficient of Gaussian discretization, (h) and (i) are corresponding residuals of real and imaginary parts.

the FWHM coefficient is too large (0.8 and 1.0), oscillations occur in the peaks resolved by the DRT, leading to distorted results. Therefore, setting the FWHM coefficient to 0.5 is an optimal choice. Fig. 3(h–i) depict the residuals of the real and imaginary parts under different FWHM coefficients, further confirming the accuracy of the results shown in Fig. 3(a).

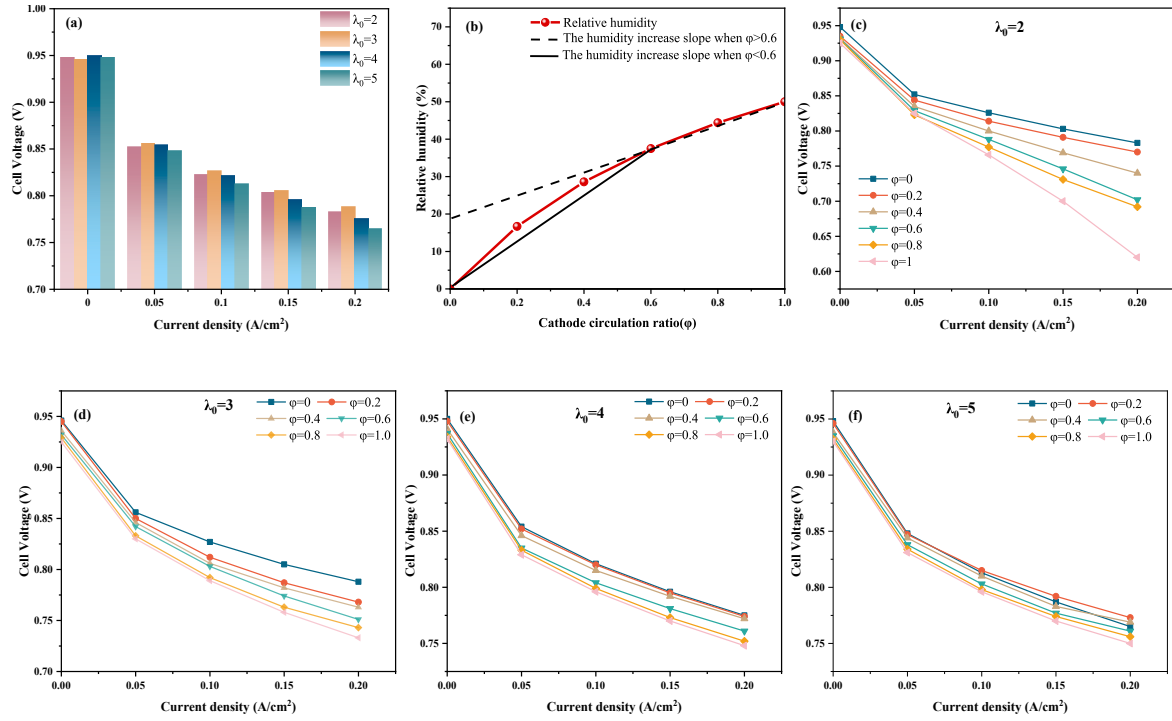
## 4. Results and discussion

### 4.1. External characteristics analysis

Fig. 4 represents the effect of cathode recirculation on the external characteristics of the PEMFC including output voltage and cathode inlet relative humidity. As shown in Fig. 4(a), the effect of fresh air stoichiometry ( $\lambda_0$ ) on the output voltage of the fuel cell in the non-recirculation mode is first explored. Four different ( $\lambda_0$ ) are analyzed in detail, which represent four different baseline cathode mass flow conditions for subsequent recirculation experiments. At each current density, the output cell voltage first rises and then falls with the increase of  $\lambda_0$ . Generally, small  $\lambda_0$  leads to insufficient supply or uneven distribution of oxygen thus affecting the output voltage, and too large  $\lambda_0$  leads to an increase in air flow rate exacerbating membrane drying inside the fuel cell thus causing a voltage drop. Therefore,  $\lambda_0 = 3$  can be considered as the best

value. Fig. 4(b) shows the change in relative humidity as  $\phi$  increases. As  $\phi$  increases, the cathode inlet relative humidity increases due to the increase in water vapor brought back from the exhaust gas. It is worth noting that the humidification ramp decreases with increasing  $\phi$ , and  $\phi = 0.6$  serves as a critical point.

Fig. 4(c–f) shows the sensitivity analysis of the current density to the  $\phi$  at different  $\lambda_0$ . At fixed  $\lambda_0$ , the voltage continues to decrease as the  $\phi$  increases, demonstrating the effectiveness of cathode recirculation on voltage suppression. The drop in output voltage becomes more sensitive to changes in  $\phi$  as the current density increases. As shown in Fig. 4(d–f), when the current density is less than 0.1 A/cm<sup>2</sup>, the  $\phi$  is increased to 2.0 before the output voltage is suppressed to within 0.8 V. This value is commonly used nowadays as a control threshold for fuel cells operating at low load conditions in order to reduce the degradation caused by high potential. Additionally, at a fixed current density, the voltage drop effect carried by the cathode recirculation ratio ( $\phi$ ) decreases as  $\lambda_0$  increases. The suppression of high potentials by cathode recirculation owes to the fact that the increase in  $\phi$  decreases the partial pressure of oxygen as well as the oxygen content inside the fuel cell. Whereas at large fresh air path stoichiometry, although the oxygen partial pressure decreases with increasing  $\phi$ , the increase in overall oxygen content offsets some of the decreasing effect. This results in a decrease in voltage drop rate for the same variation range of  $\phi$  at large  $\lambda_0$ . In order to more visually quantify



**Fig. 4.** Variation of external characteristics of PEMFC against different cathode recirculation strategy. (a) The Effect of  $\lambda_0$  on voltage under no-recirculation conditions, (b) The contribution of  $\phi$  to the relative humidity of the cathode inlet, (c)–(f) The effect of  $\phi$  on voltage at different current density under different  $\lambda_0$  conditions.

**Table 2**

Variations in voltage suppression due to  $\phi$  at different current densities and  $\lambda_0$

| Index         | $\lambda_0 = 2$ | $\lambda_0 = 3$ | $\lambda_0 = 4$ | $\lambda_0 = 5$ |
|---------------|-----------------|-----------------|-----------------|-----------------|
| $i = 0$ (OCV) | 0.023           | 0.021           | 0.018           | 0.016           |
| $i = 0.05$    | 0.027           | 0.026           | 0.025           | 0.016           |
| $i = 0.1$     | 0.06            | 0.038           | 0.025           | 0.017           |
| $i = 0.15$    | 0.103           | 0.047           | 0.026           | 0.017           |
| $i = 0.20$    | 0.163           | 0.055           | 0.027           | 0.017           |

drop

raise

the effect of the cathode recirculation, Table 2 counts the amount of output voltage suppression as the recirculation ratio is increased from 0 to 1.

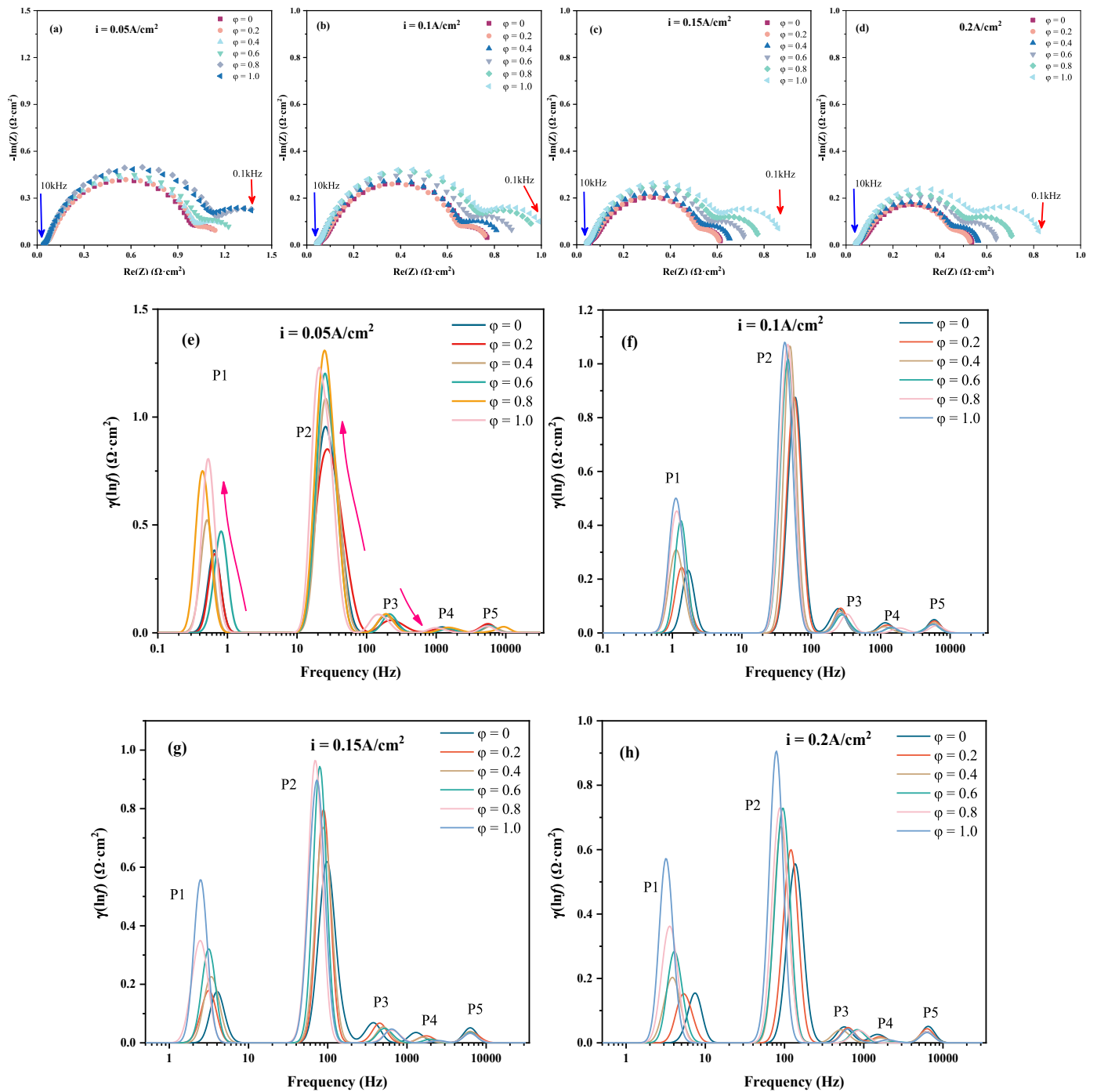
#### 4.2. Effect of cathode recirculation strategy on polarization processes

##### 4.2.1. Variation of $\phi$ at different current densities

This work conducts a series of cathode recirculation experiments at low load conditions with current density ranging from 0 to 0.2 A/cm<sup>2</sup>. To ensure measurement stability, the EIS test excludes open circuit

condition. Fig. 5 shows the Nyquist plots of the impedance test at different current densities and their corresponding DRT distributions when the fresh path air stoichiometry ratio is set to 3.0.

As shown in Fig. 5(a–c), at each current density, two distinct semi-circles can be directly observed across the impedance curve, whose radius increases and gradually shifts to the left as the cathode recirculation ratio ( $\phi$ ) increases. This indicates that the polarization resistance ( $R_p$ ) gradually increases with increasing  $\phi$  while the ohmic resistance ( $R_0$ ) has an opposite trend. The corresponding DRT distributions of polarization process are further given in Fig. 5(e–f), and it is clear that,

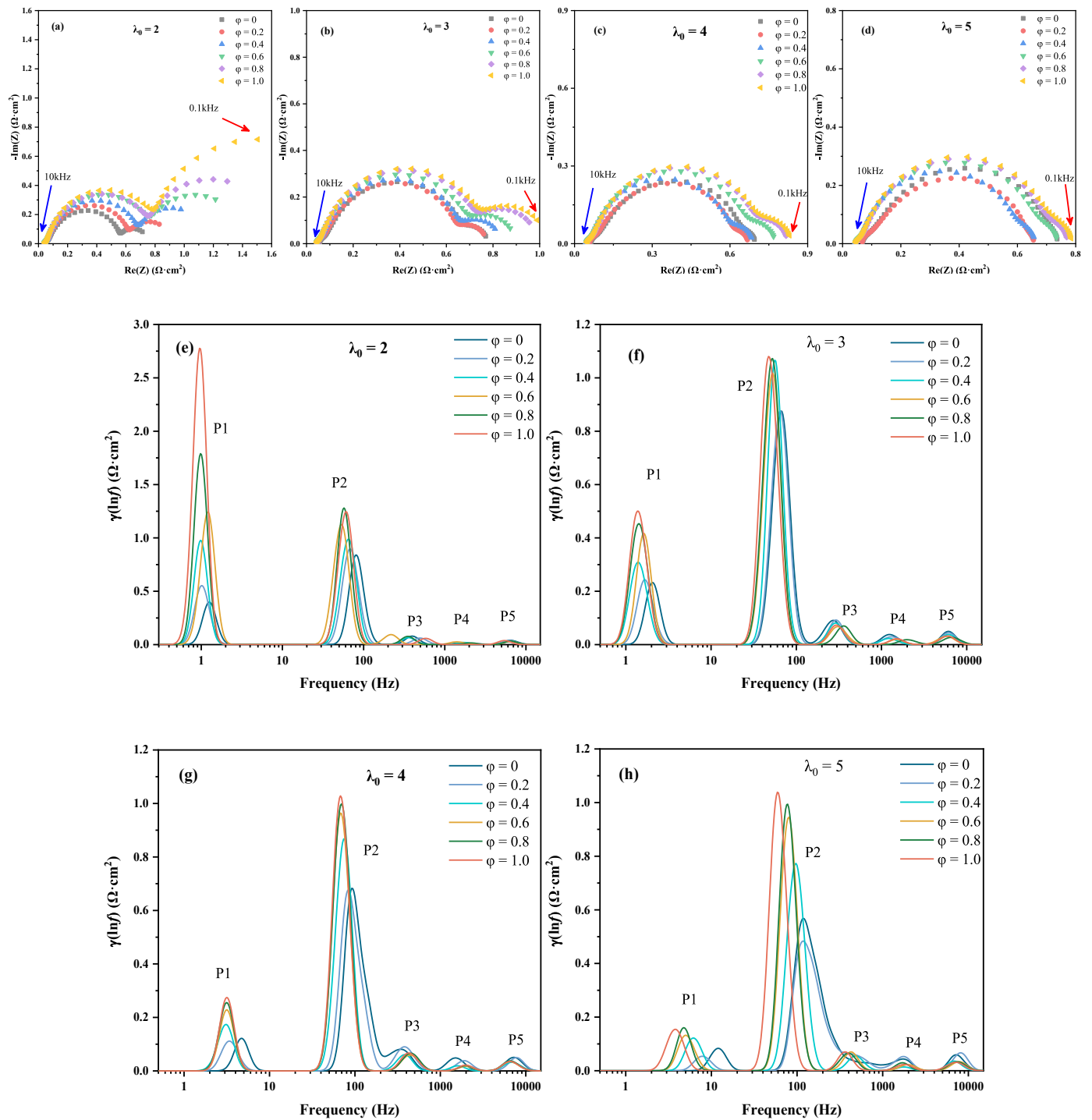


**Fig. 5.** Effect of  $\phi$  and current density on impedance spectra and polarization processes when  $\lambda = 3.0$ . (a)–(d) impedance spectra for changes in  $\phi$  at different current density; (e)–(h) corresponding DRTs.

compared to the two semicircles of the Nyquist plot, the DTR method decomposes five peaks at different eigenfrequency which indicates five different polarization dynamics processes. The distribution of the five peaks can be divided into three parts, including the low frequency (P1), the middle frequency (P2), and the high frequency (P3–P5). Under each current density, P2 dominates the polarization process, and followed by P1. This suggests that at low load conditions, oxygen reduction reaction (ORR) is the primary loss mechanism, followed by oxygen mass transfer (MT). With the gradual rise of  $\phi$  from 0 to 1, P1 and P2 exhibit an upward trend, in contrast to the gradual decline of P3–P5. This can be revealed by two effects brought about by the cathode recirculation. On the one hand, the high-frequency polarization losses in PEMFCs are

mainly related to proton transfer. An increase in the cathode recirculation ratio raises the relative humidity of the inlet gas, which consequently increases the water content of the membrane, thus decreasing the proton transfer impedance. Therefore, P1–P5 show downward trend. On the other hand, the increase in cathode recirculation ratio decreases the oxygen partial pressure as well as the oxygen content in the cathode, which increases the charge transfer impedance as well as the mass transfer impedance, and thus resulting in a continuous increase in P1 and P2. The combined effect of these two aspects explains the mechanism behind the ability of cathode recirculation to suppress high voltages.

Additionally, at fixed  $\phi$ , when the current density increases from



**Fig. 6.** Effect of  $\phi$  and  $\lambda_0$  on impedance spectra and polarization processes when  $i = 0.1 \text{ A/cm}^2$ . (a)–(d) impedance spectra for changes in  $\phi$  at different  $\lambda_0$ ; (e)–(h) corresponding DRTs.

0.05A/cm<sup>2</sup> to 0.1A/cm<sup>2</sup>, there is a significant drop both in P1 and P2. Whereas, they remain essentially unchanged with further increase in the current density. This may be attributed to the fact that at a constant stoichiometric ratio, the relatively low oxygen content at 0.05A/cm<sup>2</sup> results in a restricted oxygen reduction reaction (ORR) and a large mass transfer impedance, whereas when the current densities are in the range of 0.1A/cm<sup>2</sup> to 0.2A/cm<sup>2</sup>, the oxygen content is sufficient to support both processes smoothly.

#### 4.2.2. Variation of $\phi$ at different $\lambda_0$

In addition to the coupling of the current density and the recirculation ratio, the combination of the recirculation ratio ( $\phi$ ) and the stoichiometric ratio of the fresh air path ( $\lambda_0$ ) also plays an important role in influencing the effect of cathode recirculation. Fig. 6 shows the effect of variation of  $\phi$  and  $\lambda_0$  on the polarization process of PEMFC at current density of 0.1A/cm<sup>2</sup>. As shown in Fig. 6(a)–(d), The impedance spectra are differently affected by  $\phi$  at different  $\lambda_0$ . When  $\lambda_0 = 2$ , the radius of the second circle gradually exceeds the radius of the first circle as  $\phi$  increases. However, the first semicircle is always larger than the second

semicircle as  $\lambda_0$  increases further. At each fixed  $\varphi$ , the radius of both semicircles gradually decrease as  $\lambda_0$  increases. In addition, The trend of the impedance spectrum brought about by  $\varphi$  under each different  $\lambda_0$  is exactly the same as in Fig. 5. Accordingly, Fig. 6(e)–(f) qualitatively give the results of DRT analysis at different five eigenfrequencies. At  $\lambda_0 = 2$ , P1 gradually dominates the polarization process with respect to P2 as  $\varphi$  increases, with the inflection point at  $\varphi = 0.6$ . the oxygen diffusion impedance gradually becomes the main contributor in this case, especially at large recirculation ratios, the mass transfer impedance increases

sharply. This finding also explains why the voltage drop is more sensitive to the increase of the  $\varphi$  at small  $\lambda_0$  (as shown in Fig. 4(c)). As the  $\lambda_0$  increases, the polarization process is dominated by P2 and the eigenvalues continue to increase with the increase of  $\varphi$ . P2 is affected by charge transfer, whereas an increase in the  $\varphi$  limits the oxygen reduction reaction thereby increasing P2. Under each  $\lambda_0$ , the high-frequency part after P2 decomposes the three processes from P2 to P5, which is difficult to observe in Nyquist plot. Moreover, at fixed  $\varphi$ , the change in  $\lambda_0$  also has an effect on the polarization process. Specifically, with the increase of

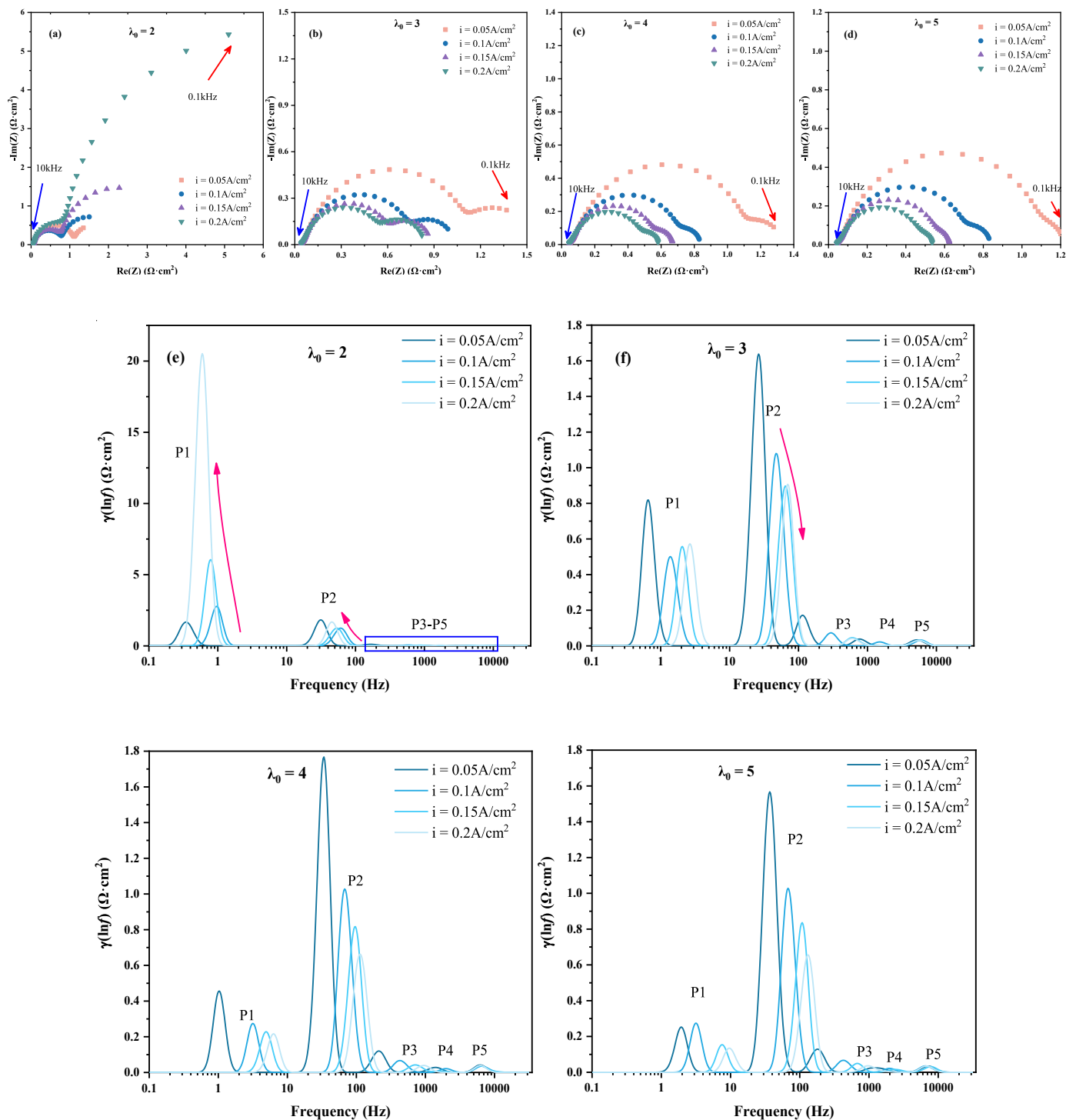


Fig. 7. Effect of  $\lambda_0$  on impedance spectra and polarization processes at different current density when  $\varphi = 1.0$ . (a)–(d) impedance spectra for changes in current density at different  $\lambda_0$ ; (e)–(h) corresponding DRTs.

$\lambda_0$ , the increase of oxygen concentration in the cathode relieves the oxygen diffusion resistance as well as promotes the ORR, which results in the gradual decrease of the two peaks P1 and P2. However, for the P3, P4 and P5 process, the change in oxygen concentration has little effect on them, as it is mainly related to the anodic reaction and proton transport.

#### 4.2.3. Effect of current density and $\lambda_0$

Fig. 7 shows the effect of changes in current density on the polarization process for different fresh air path stoichiometry conditions. In this case, the cathode recirculation ratio is set uniformly at 1.0. As shown in Fig. 7(a)–(d), the Nyquist plot shows that the ohmic internal resistance hardly varies with the current. Although an increase in current produces more water, the selected conditions are concentrated in low load conditions with limited water production, so humidification is mainly provided by the cathode recirculation. Therefore, the ohmic impedance is not sensitive to the change in current density. However, the effect of changes in current density on the polarization process is different at various  $\lambda_0$ . The limited information on the Nyquist plot can be further identified in the DRT calculations shown in Fig. 7(e)–(f). When  $\lambda_0 = 2.0$ , P1 at low frequencies dominates the polarization process and becomes more pronounced with increasing current density. This demonstrates that at  $\varphi = 1$ , a large cathode recirculation ratio greatly affects oxygen transport and diffusion at small  $\lambda_0$ , and as the current density increases, oxygen starvation becomes more severe due to the increase in oxygen consumption, resulting in a sharply larger P1. Moreover, the adjacent P3 to P5 peaks are almost negligible compared to P1 and P2. The above information also corresponds to the radius of the two arcs observed in the impedance spectrum shown in Fig. 7(a).

When  $\lambda_0$  is increased, as shown in Fig. 7(f–h), P2 completely dominates the polarization process and does not change with the change of current density. At fixed  $\lambda_0$  (3, 4 or 5), the process P2 strongly decreases with increasing current density, which is exactly opposite to the trend at small  $\lambda_0$ . This suggests that in this case, the reduction of oxygen caused by cathodic recirculation has little effect on the P2 process, since oxygen consumption is very low at such a small current density. In fact, the current dependence of P2 and its decrease at higher current densities is related to charge transfer kinetics and can be described by Ref. [39]. Furthermore, peaks P3 to P5 at high frequency have a tendency to decrease with increasing current density, which is attributed to the increase in water production at high densities reducing the proton transport resistance.

#### 4.3. Quantitative analysis of $R_0$ under different cathode recirculation strategies

The contribution of the cathode recirculation strategy to the polarization process has been assigned in detail by DRT calculations. In order to further characterize the effect of cathode recirculation on the entire

impedance inside the fuel cell, it is therefore necessary to analyze the change in ohmic resistance ( $R_0$ ). Fig. 8 demonstrates the  $R_0$  for different current densities, cathode recirculation ratios, and fresh air path stoichiometry conditions fitted by the DRT method. Specifically, Figure 8 (a) shows the variation of  $R_0$  at different current densities and  $\varphi$ . At fixed current density, the  $R_0$  shows a decreasing trend with increasing  $\varphi$ . This is attributed to the fact that the increase in the recirculation ratio increases the humidity of the cathode, which reduces the membrane impedance as well as the contact impedance between the layers. This trend also corresponds to the phenomenon of the impedance spectrum shifting to the left (in Fig. 5). However, it is evident that the rate of decrease in  $R_0$  decreases with increasing  $\varphi$ , the critical point is at  $\varphi = 0.6$ . Meanwhile, the  $R_0$  is almost constant when  $\varphi$  exceeds 1.0. At a fixed  $\varphi$ , the ohmic resistance decreases with increasing current density, which can be attributed to the increase in water production at high densities. In addition, at large  $\varphi$ , the ohmic resistance is more sensitive to changes in current. This is because the high humidity at large  $\varphi$  in combination with the high water production at high current density further reduces the ohmic resistance.

Fig. 8(b) shows the quantitative results of ohmic resistance for different  $\lambda_0$  and  $\varphi$  conditions when the current density is fixed at  $0.1 \text{ A/cm}^2$ . The phenomenon of a gradual decrease of  $R_0$  with increasing  $\varphi$  is reproduced under each  $\lambda_0$ . The amount of reduction in ohmic resistance caused by the change in  $\varphi$  is more significant for large  $\lambda_0$  conditions. Specifically, as  $\lambda_0$  increases from 2 to 5, the decrease in ohmic impedance due to the change in  $\varphi$  is  $0.006$ ,  $0.012$ ,  $0.017$ , and  $0.021 \text{ } \Omega \text{ cm}^2$ . Although the range of  $\varphi$  is the same, more water vapor is brought back at high  $\lambda_0$ , which causes more reduction in ohmic resistance. Additionally, at a fixed  $\varphi$ , the ohmic resistance gradually increases as  $\lambda_0$  increases (as shown by the black arrow). This can be attributed to the increase in flow rate, blowing the internal channels and membranes as well as the contact surfaces drier, resulting in an increase in ohmic resistance.

## 5. Conclusion

In this experiment, the performance of the cathode recirculation of a PEMFC under small load conditions is investigated. The mechanism of the cathode recirculation's influence on PEMFC performance is revealed in detail from the perspective of combining the external characteristics with the internal polarization process. The DRT method resolves the information of the electrochemical impedance spectra, which shows some obvious advantages over the ordinary Nyquist plots. The main findings of this paper are as follows:

- (1) The cathode recirculation significantly increases the relative humidity at the cathode inlet, which increases with the increase of the recirculation ratio ( $\varphi$ ), but the rate of increase gradually decreases,  $\varphi = 0.6$  can be considered as a critical point. The use of cathode recirculation for suppressing high voltages under low

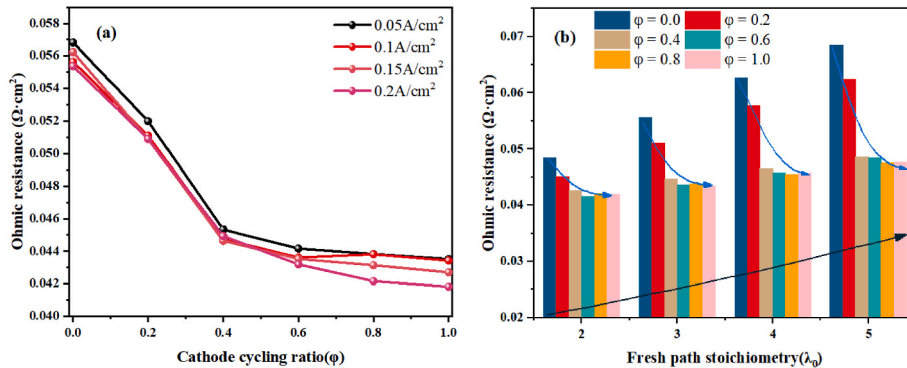


Fig. 8. Effect of cathode recirculation on ohmic resistance. (a) Variation of  $\varphi$  at different current density, (b) Variation of  $\varphi$  at different  $\lambda_0$ .

load conditions has been shown to be feasible, but in conjunction with specific strategies. The voltage drop caused by the increase in the  $\phi$  is more apparent at higher densities. Additionally, different fresh air path stoichiometries ( $\lambda_0$ ) can induce different cathode recirculation performance changes. The voltage drop due to cathode recirculation decreases with increasing  $\lambda_0$ .

- (2) Under the cathode recirculation strategy, DRT is able to separate up to five polarization processes with distinctive time constants, whereas only two arcs can be observed in the Nyquist plots. Combining the variations introduced by cathode recirculation, the low-frequency P1 process is related to oxygen diffusion, the mid-frequency P2 process is related to oxygen reduction reactions, and the remaining high-frequency processes (P3 to P5) can be assigned to proton transport processes.
- (3) As the  $\phi$  increases, the ohmic impedance ( $R_0$ ) continues to decrease, the  $R_0$  is almost constant when  $\phi$  exceeds 1.0, which explains the changes brought about by changes in humidity. The ohmic impedance continuously increase with increasing  $\lambda_0$ , and the reduction in ohmic impedance under higher  $\lambda_0$  is greater for the same range of variation of  $\phi$ .

### CRediT authorship contribution statement

**Ze Liu:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mingyang Yang:** Validation, Investigation. **Sichuan Xu:** Writing – review & editing, Supervision, Resources, Conceptualization.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

Data will be made available on request.

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### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2024.234165>.

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